

Fixed Points II: Projected Fixed Points and Equilibria in a 10D Modal Model

Peter Nero

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Version 6

Abstract

We apply the canonical FP–I machinery to a ten-dimensional control setting $M_{10} = Y^4 \times X^6$. The compact six-manifold X^6 carries three compatible vertical structures represented by strongly commuting nonnegative self-adjoint operators. Overlap is allowed; nesting requires supplied inclusion maps and is not inferred from ranks $1 < 2 < 3$. Shared phase data may be represented by a common unitary line connection; this does not add a seventh internal product dimension. We prove existence of projected time- τ fixed points by applying the canonical FP–I Schauder/Darbo gates to this ten-dimensional operator, and prove coherent uniqueness under base coercivity or strong monotonicity. A projected fixed point is promoted to a full equilibrium only under a strict Lyapunov identity. Fiber gaps control only the noncoherent Q sector and are never used as coherent damping.

Revision note for version 6

Supersedes. Version 5, retained as the released edition.

Reason. The analytic development was independent of later research, but an application-specific carrier remark cited a downstream MTT repository.

Resolution. The remark is replaced by a self-contained tensor-factor example. All mathematical imports are now from FP I or standard literature; application-specific source selection is outside this foundational paper.

Retained result. The joint-projector, existence, coherent contraction and equilibrium-promotion statements retain their explicit hypotheses.

Remaining boundary. A concrete application must provide its operators, domains, energy and interpretation of the control parameter.

Revision note for version 5

Supersedes. *Fixed Points II: Projected Fixed Points and Equilibria in a 10D Modal Model*, version 4.

Reason. Version 4 gives the corrected ten-dimensional application, but the relation between complement damping, coherent contraction, and equilibrium promotion remained too compressed for a standalone reading.

Resolution. Version 5 adds a paper-specific reading guide and examples that separate those three mechanisms, explains the lane factor and shared line without adding dimensions,

and keeps the generic fixed-point theorems owned by Fixed Points I. Publication description, revision history, and computational provenance are now separate.

Retained result. The version 4 Schauder/Darbo application, conditional coherent uniqueness, and strict-Lyapunov equilibrium gate are unchanged.

Remaining boundary. Physical q79 HYM endpoints, action, Hessian, finite reduction, Lorentzian time, and observable interpretation remain open downstream inputs.

Revision note for version 4

Supersedes. *Fixed Points II: Projected Fixed Points and Equilibria in a 10D Modal Model*, version 3.

Reason. The version 3 correction fixed the ten-dimensional control geometry but still described the q79 $1 < 2 < 3$ carrier too loosely. The exact projective-module theorem excludes placing a nontrivial parallel flag inside an irreducible stable HYM factor and distinguishes post-projection finite algebra from a physical Galerkin subspace.

Resolution. Version 4 retains the corrected FP theorem and places the rank flag on an external lane tensor factor, with the shared differential line as a separate flat scalar factor. It also records that the 27-state algebra is post-projection data and that the six-coordinate strain carrier is a nonlinear quotient shadow rather than a linear subspace of the physical HYM carrier. Generic existence and equilibrium-promotion results remain owned by FP-I; this paper states their ten-dimensional application corollaries so that it is standalone without duplicating the generic proofs.

Retained result. Schauder/Darbo existence and conditional coherent uniqueness survive in the corrected ten-dimensional control realization.

Remaining boundary. The selected visible/hidden q79 HYM endpoints, physical action and Hessian, finite invariant subspace or Feshbach execution, physical time, and Lorentzian dynamics require independent source and completion theorems.

How to read this application

FP-I proves the generic analytic machinery. This paper asks what that machinery becomes in a ten-dimensional control model with one compact six-dimensional internal space and three compatible vertical operators. Its formal statements are therefore application corollaries and model-specialized estimates, not replacement copies of the FP-I theorems.

The central picture in plain language. There are two independent stabilization questions. First, does the vertical operator suppress every component outside the joint harmonic sector? This is controlled by the fiber gaps and concerns $Q\Psi$. Second, does motion inside the joint harmonic sector contract to one state? The vertical operators vanish there, so this requires a base Poincaré gap or strong monotonicity of the coherent nonlinearity. Confusing these mechanisms would make a fiber gap appear to prove uniqueness when it only damps discarded internal modes.

Argument map. Section 1 states the ten-dimensional control scope. Section 2 places the three operators on the same internal Hilbert space, constructs their joint projector, and gives a shared-line tensor-factor example. Section 3 then separates the Π_{coh} and Q evolutions,

promotes a projected return only under a strict Lyapunov identity, gives the two coherent contraction routes, and transfers the compact-base and noncompact-base existence gates from FP–I. Section 4 supplies examples of the operator hypotheses. The appendices provide the specialized contraction and damping calculations.

What this paper does not select. The notation $M_{10} = Y^4 \times X^6$ is a control realization, not a proof that physical spacetime is this Riemannian product. Nor does the paper derive a physical bundle, identify the three operators with finite rank-flag projectors, or derive a Galerkin basis from rank counting. A physical application must construct those objects and verify the present hypotheses; its construction is not a premise imported into this paper.

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1 Introduction and scope

We consider $M_{10} = Y^4 \times X^6$, where (Y^4, g_Y) is a complete Riemannian analytic/control manifold and X^6 is compact. The Riemannian base may model a Cauchy-slice control geometry or Euclideanized problem, but is not identified here with physical Lorentzian spacetime. The internal space carries three vertical structures indexed by $n = 1, 2, 3$, each equipped with a *nonnegative* self-adjoint vertical operator and a uniform spectral gap. We adopt throughout the corrected FP-I series conventions:

- Laplacians are *nonnegative* ($\Delta \geq 0$), and dissipative flows are written $\partial_t \Psi = -A\Psi - N(\Psi)$.
- The coherent projector Π_{coh} is the joint fiber-harmonic projector, and $Q := \text{Id} - \Pi_{\text{coh}}$.
- Semigroup smoothing is recorded in the global safe form $\|A^{1/2}e^{-tA}Q\| \lesssim (1+t^{-1/2})e^{-\lambda t}$ (cf. analytic semigroup bounds in [1, 4]).

This alignment is what allows FP-III and later papers to reuse damping/smoothing constants without sign drift.

The parameter t in the parabolic flow and the step τ below are stabilization parameters. A physical Lorentzian evolution $U(t_2, t_1)$ and any relation to this control flow require a separate theorem.

Our goals are: (i) existence of coherent fixed points via Schauder (compact base) or Darbo (noncompact base with confinement); (ii) uniqueness via a Fundamental Contractivity Condition (FCC) on the coherent sector; (iii) explicit constants in standard geometries. General dynamical-systems context may be compared with [7].

We also cite here FP-I as the foundational reference for the projection/fixed-point strategy: see [6].

2 Geometry and functional setup

2.1 Internal bundles and harmonic alignment

Let X^6 be a compact six-manifold. On the common internal Hilbert space $L^2(X^6)$ let $A_n(y)$, $n = 1, 2, 3$, denote the nonnegative self-adjoint operators associated with the selected vertical structures. These structures may overlap. They are nested only if a concrete realization supplies bundle injections or operator intertwiners; rank labels alone do not do so. No product decomposition into three disjoint fibers is assumed. The shared central circle is encoded by a common circle cycle, foliation, or $U(1)$ line-bundle connection on X^6 . Its line-bundle data do not add the dimension of the bundle total space to $\dim X^6$.

Remark 2.1 (A tensor-factor example, not a selected realization). Let $\mathcal{H}_0$ be a Hilbert space and let L_y be a one-dimensional Hermitian fiber over the base. On

$$\mathcal{H}_0 \otimes \mathbb{C}^3 \otimes L_y$$

the matrices $p_1 = \text{diag}(1, 0, 0)$, $p_2 = \text{diag}(1, 1, 0)$, and $p_3 = I_3$ act only on $\mathbb{C}^3$, by tensoring with the identity on the other factors. Direct multiplication gives $p_i p_j = p_{\min(i,j)}$. Scalar phase multiplication on L_y commutes with all three. Thus a common phase and an increasing finite rank flag can coexist without extra base dimensions. This example neither selects such a factorization nor identifies p_n with the harmonic projectors of the differential operators below. Those operators, their gaps, and their commutation remain independent hypotheses. A finite fiber rank is not a finite-dimensional space of sections.

We assume bounded internal geometry and smooth bounded dependence on $y \in Y^4$ as in FP-I. A concrete MTT realization must specify X^6 , the three operators, their domains, and the shared-circle action.

Assumption 2.2 (Strong commutation of vertical operators). For every y , the spectral measures of $A_1(y), A_2(y), A_3(y)$ commute. Their common domain is dense, their weighted form sum is closed, and the commutation and domain bounds are uniform in y . Equivalently, the three operators admit a joint spectral calculus on the common internal Hilbert space.

2.2 Fiber Laplacians and gaps

Write $A_n(y)$ for the *nonnegative* vertical Laplace-type operator (or the vertical part of the spinor Laplacian in the spinor option). Each $A_n(y)$ has discrete spectrum with 0 the harmonic eigenvalue. We assume a uniform gap

$$\lambda^* := \inf_{y \in Y^4} \min_{1 \leq n \leq 3} \lambda_1(A_n(y)) > 0, \quad (2.1)$$

and constant harmonic rank in y .

Explicit gaps in standard fibers. For S_ℓ^1 (length ℓ), $\lambda_1 = (2\pi/\ell)^2$. For a flat torus T^m with side lengths L_j , $\lambda_1 = (2\pi)^2 \min_j L_j^{-2}$. For the round S^3 (radius 1), $\lambda_1 = 3$ (see, e.g., [2]). Nilmanifold factors with bounded geometry admit a uniform positive lower bound on λ_1 once the metric class is fixed (bounded geometry prevents collapse).

2.3 Fields, norms, and (optional) spinors

We work with complex scalar fields by default; all results extend to sections of a fixed Hermitian bundle with compatible connection.

Anisotropic norm. Let

$$H_F^1 := \bigcap_{n=1}^3 \text{dom}(A_n^{1/2}), \quad \|\Psi\|_{H_F^1}^2 := \|\Psi\|_{L^2}^2 + \sum_{n=1}^3 \|A_n^{1/2} \Psi\|_{L^2}^2,$$

with the natural common form domain. This is the joint vertical anisotropic norm for the possibly overlapping structures. Under strong commutation it is equivalent to the form norm of the weighted vertical sum when the weights κ_n are bounded above and below away from zero.

2.4 Joint harmonic projector

Let $\Pi_n(y)$ be the L^2 -orthogonal projector onto $\ker A_n(y)$ and define the joint projector fiberwise by

$$\Pi_{\text{coh}}(y) := \Pi_1(y) \Pi_2(y) \Pi_3(y).$$

We view Π_{coh} as the induced projector on $L^2(M_{10})$ acting fiberwise.

Lemma 2.3 (Commutation and range). *Under Assumption 2.2, the spectral projectors commute. Hence*

$$\Pi_{\text{coh}}(y) = \Pi_1(y) \Pi_2(y) \Pi_3(y), \quad \text{Ran } \Pi_{\text{coh}}(y) = \bigcap_{n=1}^3 \ker A_n(y).$$

Geometric meaning of the joint projector. Each Π_n asks whether a field is harmonic for one selected vertical operator. Their product retains only fields that answer yes to all three questions at once. Strong commutation is what makes this simultaneous test a genuine orthogonal projector rather than an order-dependent sequence of filters. Overlap of the underlying vertical structures is harmless at this operator level; literal nesting still requires independently supplied maps.

Corollary 2.4 (10D joint-projector Sobolev regularity). *Under the gap condition (2.1) and bounded fiber geometry, the joint projector $\Pi_{\text{coh}} : H^1 \rightarrow H^1$ is bounded with*

$$\|\Pi_{\text{coh}}\|_{H^1 \rightarrow H^1} \leq C_\Pi,$$

and $\text{Ran } \Pi_{\text{coh}}$ is closed in H^1 .

This is the commuting three-projector specialization of the FP–I Sobolev projector theorem [6]; the Riesz–Dunford and elliptic regularity hypotheses are stated here explicitly.

Remark 2.5 (Reference for Riesz projector bounds). Uniform control of fiberwise Riesz projectors on Sobolev scales follows from the Riesz–Dunford formula and parameter-dependent elliptic regularity; see [5, 3].

3 Flow, projection, and fixed points

3.1 Parabolic generator and smoothing

Let $\Delta_Y \geq 0$ be the Riemannian Laplacian of the base control geometry (or the Laplacian on a selected Riemannian Cauchy slice with declared boundary conditions). It is not the Lorentzian d’Alembertian. We study

$$\partial_t \Psi = -\left(\sum_{n=1}^3 \kappa_n A_n + \varepsilon \Delta_Y\right) \Psi - N(\Psi), \quad \kappa_n > 0, \varepsilon \in [0, 1]. \quad (3.1)$$

Denote the time- τ map by Φ_τ and set

$$Q := \text{Id} - \Pi_{\text{coh}}, \quad A := \sum_{n=1}^3 \kappa_n A_n + \varepsilon \Delta_Y.$$

Define rates

$$\lambda_n := \inf_{y \in Y^4} \lambda_1(A_n(y)) > 0, \quad \lambda_A := \min_{1 \leq n \leq 3} \kappa_n \lambda_n.$$

Under strong commutation, λ_A is a lower bound for the vertical form sum on $\text{Ran } Q$.

Assumption 3.1 (Projector commutation for the base-regularized model). The common domain is preserved by Π_{coh} and $[A, \Pi_{\text{coh}}] = 0$. If the vertical operators depend on y , this includes control of the commutator $[\Delta_Y, \Pi_{\text{coh}}]$; it is not automatic from strong commutation of the A_n . Models with nonzero commutator require explicit leakage terms and are outside the exact invariant-sector theorem below.

By analytic semigroup theory (e.g. [1, 4]) and Assumption 3.1, for $t > 0$ one has

$$\|A^{1/2} e^{-tA} Q\|_{L^2 \rightarrow L^2} \leq C(1 + t^{-1/2}) e^{-\lambda_A t}, \quad \|e^{-tA} \Pi_{\text{coh}}\|_{L^2 \rightarrow L^2} \leq 1. \quad (3.2)$$

For two solutions let $w = \Psi_1 - \Psi_2$. Duhamel’s formula gives the sector-correct estimate

$$\|Qw(t)\|_{H_F^1} \leq C(1 + t^{-1/2}) e^{-\lambda_A t} \|Qw(0)\|_{L^2}$$

$$+ C \int_0^t (1 + (t-s)^{-1/2}) e^{-\lambda_A(t-s)} \|Q(N(\Psi_1(s)) - N(\Psi_2(s)))\|_{L^2} ds. \quad (3.3)$$

No factor $e^{-\lambda_A t}$ is asserted for $\Pi_{\text{coh}} w$. On the invariant coherent sector its dynamics are instead

$$\partial_t u = -\varepsilon \Delta_Y u - \Pi_{\text{coh}} N(u), \quad u = \Pi_{\text{coh}} u, \quad (3.4)$$

and require base coercivity or coherent monotonicity for contraction.

Why the sector split matters. Equation (3.3) says that positive vertical modes are damped up to nonlinear forcing. Equation (3.4) says that harmonic modes do not feel that vertical gap at all. Thus approaching the coherent sector and selecting a unique point inside it are two different mathematical events. The first can occur while an entire family of coherent states remains.

Remark 3.2 (Base lower-order terms). If the Q equation includes first/zero-order terms with relative bound $C_Y \geq 0$, replace the noncoherent decay margin by $\lambda_A - C_Y$. This modification remains a Q -sector statement.

3.2 Coherence invariance and the projected map

Assumption 3.3 (Coherence invariance). We assume Assumption 3.1 and $N(\text{Ran } \Pi_{\text{coh}}) \subset \text{Ran } \Pi_{\text{coh}}$.

We apply fixed-point principles to

$$T_\tau := \Pi_{\text{coh}} \circ \Phi_\tau : D \rightarrow \text{Ran } \Pi_{\text{coh}} \cap L^2,$$

where $D \subset \text{Ran } \Pi_{\text{coh}} \cap L^2$ is a closed bounded set (specified in §3.4).

Assumption 3.4 (Strict Lyapunov identity). There is a functional $\mathcal{E}$ bounded below on the invariant set such that every sufficiently regular trajectory satisfies

$$\mathcal{E}(\Phi_t u) + c \int_0^t \|\partial_s \Phi_s u\|_{L^2}^2 ds = \mathcal{E}(u), \quad c > 0.$$

For a gradient flow this follows from $N = \nabla V$ with the required domain and regularity. It is not implied by semilinearity alone.

Corollary 3.5 (10D projected-step equilibrium promotion). *Under Assumption 3.3, a point $\Psi^* \in \text{Ran } \Pi_{\text{coh}}$ with $T_\tau(\Psi^*) = \Psi^*$ is a fixed point of the time- τ stabilization map. If Assumption 3.4 also holds, then it is a genuine equilibrium: $\Phi_t(\Psi^*) = \Psi^*$ for all $t \geq 0$.*

Proof. Apply the FP-I promotion theorem [6]. In the present realization, coherence invariance gives $T_\tau(\Psi^*) = \Phi_\tau(\Psi^*) = \Psi^*$. The Lyapunov identity on $[0, \tau]$ then has equal endpoint energies, so its nonnegative dissipation integral vanishes. Hence $\partial_t \Phi_t(\Psi^*) = 0$ and the orbit is stationary. Without the Lyapunov identity, a time- τ fixed point could be a nonstationary periodic point. $\square$

What equilibrium promotion establishes. The corollary closes a logical gap, not a dynamical one. A projected time- τ return becomes an equilibrium because strict energy loss forbids a nonstationary loop with equal endpoint energy. It does not prove that a return exists or that it is unique; those are the separate existence and FCC questions below.

3.3 Uniqueness via the FCC (anisotropic norm)

Theorem 3.6 (Fundamental contractivity on the coherent sector). *Let*

$$L_{\text{coh}} := \text{Lip}(\Pi_{\text{coh}} N|_{\text{Ran } \Pi_{\text{coh}}}).$$

Then for $\Psi_1, \Psi_2 \in \text{Ran } \Pi_{\text{coh}}$,

$$\|\Phi_\tau(\Psi_1) - \Phi_\tau(\Psi_2)\|_{L^2} \leq e^{L_{\text{coh}}\tau} \|\Psi_1 - \Psi_2\|_{L^2},$$

and since $\|\cdot\|_{H_F^1} = \|\cdot\|_{L^2}$ on $\text{Ran } \Pi_{\text{coh}}$, the same holds in H_F^1 . Moreover, $T_\tau = \Pi_{\text{coh}} \Phi_\tau$ is a Banach contraction on the declared coherent phase space provided either:

- (a) **Base diffusion:** $\varepsilon > 0$ and the nonnegative Δ_Y has a Poincaré gap $\mu_Y > 0$ on a specified invariant subspace (for example mean-zero functions or Dirichlet boundary data), with $\varepsilon\mu_Y > L_{\text{coh}}$; then

$$\|T_\tau(\Psi_1) - T_\tau(\Psi_2)\|_{L^2} \leq e^{-(\varepsilon\mu_Y - L_{\text{coh}})\tau} \|\Psi_1 - \Psi_2\|_{L^2}.$$

- (b) **Strong monotonicity:** there exists $\mu > 0$ such that $\langle \Pi_{\text{coh}} N(u) - \Pi_{\text{coh}} N(v), u - v \rangle_{L^2} \geq \mu \|u - v\|_{L^2}^2$ for all $u, v \in \text{Ran } \Pi_{\text{coh}}$; then

$$\|T_\tau(\Psi_1) - T_\tau(\Psi_2)\|_{L^2} \leq e^{-\mu\tau} \|\Psi_1 - \Psi_2\|_{L^2}.$$

Remark 3.7 (Base zero mode). On a compact boundaryless connected base, scalar constants lie in $\ker \Delta_Y$. Therefore base diffusion gives no coherent contraction on the full scalar space. One must remove/fix the mean, impose boundary conditions, or use coherent strong monotonicity.

Remark 3.8 (Why fiber smoothing does not contract on $\text{Ran } \Pi_{\text{coh}}$). On $\text{Ran } \Pi_{\text{coh}}$ one has $\nabla_F \Psi = 0$, hence $\|\Psi\|_{H_F^1} = \|\Psi\|_{L^2}$ and $Q = 0$. Thus contraction must come from *base* dissipation (case (a)) or coherent monotonicity (case (b)), not from the fiber semigroup estimate (3.2).

Interpretation of the FCC. The FCC is a uniqueness gate. Its contraction factor compares two coherent initial states after one stabilization step. Base diffusion supplies that factor only after its zero mode has been removed or fixed; strong monotonicity supplies it directly from the nonlinear response. Either route can make Banach's theorem available, but neither is a consequence of the internal spectral gap.

3.4 Existence without contraction: Schauder and Darbo

Uniqueness is useful when a contraction is available, but existence should not be made to depend on it. On a compact base, smoothing and Rellich compactness lead to Schauder. On a noncompact base, confinement replaces global compactness by the compact-local/small-tail decomposition inherited from FP-I. The invariant set is part of either theorem and must be constructed in the concrete model.

Assumption 3.9 (Confinement for noncompact bases). If Y^4 is noncompact, assume the energy includes a confining base potential $V(y) \rightarrow \infty$ as $r(y) := \text{dist}_Y(y, y_0) \rightarrow \infty$, yielding uniform tail control on energy sublevels (as in FP-I's confinement hypothesis).

Assumption 3.10 (Base smoothing for compactness). For the Schauder/Darbo existence routes we assume *base smoothing* is present, e.g. $\varepsilon > 0$ in (3.1) (or one works with a base-regularized generator as in FP-I). This ensures Φ_τ maps bounded sets in L^2 into sets bounded in H^1 on compact base regions, so Rellich compactness applies.

Lemma 3.11 (Coherent affine bound). *Assume one of the two contraction hypotheses in Theorem 3.6, with coherent contraction rate*

$$\rho_{\text{coh}} = \begin{cases} \varepsilon\mu_Y - L_{\text{coh}}, & \text{base-gap case,} \\ \mu, & \text{strong-monotonicity case.} \end{cases}$$

Set $q_{\text{coh}}(\tau) = e^{-\rho_{\text{coh}}\tau}$ and $B_{\text{coh}}(\tau) = \|T_\tau(0)\|_{L^2}$. Then

$$\|T_\tau(\Psi)\|_{L^2} \leq q_{\text{coh}}(\tau)\|\Psi\|_{L^2} + B_{\text{coh}}(\tau)$$

for coherent Ψ . No Q -sector gap enters this estimate.

Corollary 3.12 (Invariant/absorbing ball). *If $q_{\text{coh}}(\tau) < 1$ in Lemma 3.11, then*

$$D_R := \{\Psi \in \text{Ran } \Pi_{\text{coh}} \cap L^2 : \|\Psi\|_{L^2} \leq R\}, \quad R \geq \frac{B_{\text{coh}}(\tau)}{1 - q_{\text{coh}}(\tau)},$$

is forward invariant for T_τ and absorbing in the coherent L^2 metric.

Corollary 3.13 (Compact-base FP-I gate in the 10D model). *Assume Y^4 is compact and Assumption 3.10 holds. Then $T_\tau = \Pi_{\text{coh}}\Phi_\tau$ maps bounded L^2 sets into sets relatively compact in L^2 . If a nonempty $D \subset \text{Ran } \Pi_{\text{coh}} \cap L^2$ is L^2 -closed, bounded, convex, and satisfies $T_\tau(D) \subset D$, then T_τ has a fixed point in D .*

Lemma 3.14 (10D compact-local/small-tail reduction). *Assume Assumptions 3.10 and 3.9. Let D_M be a forward-invariant energy sublevel that is bounded in H^1 and has the uniform tail control of Assumption 3.9. Let $\alpha(\cdot)$ be the Kuratowski measure of noncompactness in L^2 . Then Φ_τ is Sadoskii-condensing on D_M : for every bounded $E \subset D_M$ with $\alpha(E) > 0$,*

$$\alpha(\Phi_\tau(E)) < \alpha(E).$$

Transfer of the FP-I mechanism. Fix a bounded set E in a forward-invariant energy sublevel. Let $\chi_R \in C_c^\infty(Y^4)$ be a cutoff equal to 1 on $\{r \leq R\}$ and supported on $\{r \leq 2R\}$, and extend χ_R to M_{10} by $\chi_R \circ \pi$. Define the output decomposition

$$K_R(\Psi_0) := (\chi_R \circ \pi) \Phi_\tau(\Psi_0), \quad S_R(\Psi_0) := (1 - \chi_R \circ \pi) \Phi_\tau(\Psi_0).$$

Confinement gives uniform small tails: $\sup_{\Psi_0 \in E} \|S_R(\Psi_0)\|_{L^2} \rightarrow 0$ as $R \rightarrow \infty$. Base smoothing (Assumption 3.10) implies that for each fixed R , the set $K_R(E)$ is relatively compact in L^2 (Rellich on the compact base region $\{r \leq 2R\}$ plus smoothing in H^1). Hence $\alpha(K_R(E)) = 0$, and by standard properties of α ,

$$\alpha(\Phi_\tau(E)) = \alpha(K_R(E) + S_R(E)) \leq \alpha(K_R(E)) + 2 \sup_{\Psi_0 \in E} \|S_R(\Psi_0)\|_{L^2} = 2 \sup_{\Psi_0 \in E} \|S_R(\Psi_0)\|_{L^2}.$$

Choose R so large that the right-hand side is $< \alpha(E)$, giving $\alpha(\Phi_\tau(E)) < \alpha(E)$. $\square$

Corollary 3.15 (Noncompact-base FP-I gate in the 10D model). *Assume Y^4 is noncompact and Assumptions 3.9 and 3.10 hold. Let $D \subset \text{Ran } \Pi_{\text{coh}} \cap L^2$ be nonempty, L^2 -closed, bounded, convex, and invariant under T_τ . Since Π_{coh} is an L^2 -orthogonal projection of norm one, Lemma 3.14 implies that T_τ is condensing on D . Hence T_τ has a fixed point in D .*

Remark 3.16 (No full-map decay from the fiber gap). The margin $\lambda_A - C_Y - L_{\text{Lip}}$ can sharpen estimates for the Q component. It cannot imply $\alpha(\Phi_\tau(E)) \leq \theta(\tau)\alpha(E)$ with $\theta(\tau) \rightarrow 0$ for the full map unless the coherent component has its own base or monotonicity contraction.

What the existence gates contribute. The compact and condensing routes answer only whether T_τ has at least one coherent fixed point in the declared invariant set. Combined with the strict Lyapunov identity, such a point is a full equilibrium. Combined further with the FCC, it is the unique equilibrium in the relevant coherent phase space. Keeping these three gates separate makes clear which conclusion fails when compactness, Lyapunov strictness, or coherent contraction is absent.

3.5 Damping estimates and effective rates

Let $\Psi = \Pi_{\text{coh}}\Psi + Q\Psi$ and write $\Psi^\perp := Q\Psi$. Assuming coherence invariance and that N is L_{Lip} -Lipschitz on the invariant set considered, compare Ψ with $\Pi_{\text{coh}}\Psi$ to obtain

$$\frac{d}{dt}\|\Psi^\perp\|_{L^2}^2 \leq -2\lambda_A\|\Psi^\perp\|_{L^2}^2 + 2C_Y\|\Psi^\perp\|_{L^2}^2 + 2L_{\text{Lip}}\|\Psi^\perp\|_{L^2}^2.$$

Hence the effective decay rate satisfies

$$\eta \geq \lambda_A - C_Y - L_{\text{Lip}}. \quad (3.5)$$

This governs noncoherent L^2 decay only. Higher-norm control follows from the sector-correct Duhamel estimate (3.3) under its regularity hypotheses. It does not imply coherent or full H^1 decay.

4 Examples

4.1 Product of circles and tori

Let $X^6 = T_1^2 \times T_2^2 \times T_3^2$. Let A_n contain the nonnegative Laplacian in the T_n^2 directions, and let a common $U(1)$ connection encode the selected central-circle holonomy without adding another product factor. For rectangular tori,

$$\lambda_n = \min\left\{(2\pi/\ell_{u_n})^2, (2\pi/\ell_{v_n})^2\right\} > 0,$$

for the first nonconstant mode of that factor. These gaps control Q ; the FCC margin still requires base coercivity or coherent monotonicity.

4.2 Sphere case (S^3)

If one selected vertical operator has round- S^3 leaves of radius one, its first positive eigenvalue is 3 (see [2]). This is an operator-level gap example, not a claim that three disjoint S^3 factors fit inside X^6 or that the coherent sector contracts.

4.3 Nilmanifold auxiliary model

On a compact six-manifold carrying a nilmanifold foliation or on an auxiliary Lens–Nil model carrying a nilpotent vertical operator, a noncollapsing bounded-geometry family has a uniform positive first nonzero eigenvalue once its metric class is fixed. The actual embedding, shared-circle action, and commutation with the other two operators must be checked in the concrete model. This example does not identify the auxiliary model with a uniquely selected physical compactification.

4.4 Toward Physical Interpretation (programmatic note)

While this paper is purely mathematical, the following schematic links guide later work:

- The three strongly commuting operators encode compatible, possibly overlapping modal structures on the same internal space.
- The coherent sector $\text{Ran } \Pi_{\text{coh}}$ consists of fiberwise zero modes (only base dependence remains).
- Spectral gaps λ_n set decay rates of nonzero fiber modes; in KK heuristics this corresponds to mass scales $m_n \sim \lambda_n^{1/2}$ for excitations.
- A projected step fixed point is an equilibrium only when the strict Lyapunov hypothesis of Proposition 3.5 is verified.

5 Concluding remarks

We specialized the FP–I framework to $M_{10} = Y^4 \times X^6$ with three strongly commuting vertical operators on one compact internal space. Their joint spectral projector gives a precise coherent sector, vertical gaps damp its complement, and independent base or monotonicity hypotheses control contraction inside it. Compact-base Schauder and noncompact-base Darbo gates provide alternative existence mechanisms; a strict Lyapunov identity then distinguishes an actual equilibrium from a merely periodic projected return.

The useful result is the resulting dependency chain:

$$\begin{aligned} \text{joint operator data} &\implies \text{coherent projector} \implies \text{existence}, \\ \text{existence} &\implies \text{equilibrium promotion} \implies \text{conditional uniqueness}. \end{aligned}$$

No arrow may be skipped. In particular, the fiber gap is not a coherent contraction constant, the shared circle is not an extra internal dimension, and a finite rank flag does not specify the vertical differential operators.

What remains is a realization problem rather than another abstract fixed-point lemma: a selected physical model must provide the vertical operators, their common domains and commutators, the physical action and strict Lyapunov functional, and the relation between stabilization and Lorentzian evolution. Until those data are supplied, this paper establishes a conditional ten-dimensional control theorem, not a derived physical vacuum.

A Contraction on the coherent sector: details

Throughout, let $T_\tau = \Pi_{\text{coh}} \Phi_\tau$ act on the declared coherent phase space and write $L_{\text{coh}} := \text{Lip}(\Pi_{\text{coh}} N|_{\text{Ran } \Pi_{\text{coh}}})$. Coherence invariance means $[A, \Pi_{\text{coh}}] = 0$ and $\Pi_{\text{coh}} N(\text{Ran } \Pi_{\text{coh}}) \subset \text{Ran } \Pi_{\text{coh}}$.

Proof of Theorem 3.6 (baseline). If $\Psi_i(0) \in \text{Ran } \Pi_{\text{coh}}$, then $\Phi_t(\Psi_i) \in \text{Ran } \Pi_{\text{coh}}$ for all $t \geq 0$. Set $u(t) := \Phi_t(\Psi_1) - \Phi_t(\Psi_2) \in \text{Ran } \Pi_{\text{coh}}$. Then $\partial_t u = -\Pi_{\text{coh}}(N(\Phi_t(\Psi_1)) - N(\Phi_t(\Psi_2)))$, hence $\frac{d}{dt} \|u\|_{L^2} \leq L_{\text{coh}} \|u\|_{L^2}$ and Grönwall yields the bound. $\square$

Proof of Theorem 3.6(a) (base diffusion). Assume $\varepsilon > 0$ and the nonnegative Δ_Y has a Poincaré gap $\mu_Y > 0$ on the declared mean-zero or boundary-conditioned coherent subspace. On that subspace the linear generator is $-\varepsilon \Delta_Y$, so for $u(t)$ as above,

$$\frac{d}{dt} \|u\|_{L^2}^2 \leq -2(\varepsilon \mu_Y - L_{\text{coh}}) \|u\|_{L^2}^2,$$

and Grönwall gives the stated contraction. $\square$

Proof of Theorem 3.6(b) (strong monotonicity). Assume $\langle \Pi_{\text{coh}} N(u) - \Pi_{\text{coh}} N(v), u - v \rangle_{L^2} \geq \mu \|u - v\|_{L^2}^2$ on $\text{Ran } \Pi_{\text{coh}}$. Then $\frac{d}{dt} \|u\|_{L^2}^2 \leq -2\mu \|u\|_{L^2}^2$ and Grönwall yields the contraction. $\square$

B Spectral gap and damping inequality

Let $\Psi = \Pi_{\text{coh}} \Psi + Q\Psi$ and $\Psi^\perp := Q\Psi$. For solutions of (3.1), using that $A \geq \lambda_A I$ on $\text{Ran } Q$ and that N is L_{Lip} -Lipschitz on the invariant set, one obtains

$$\frac{d}{dt} \|\Psi^\perp(t)\|_{L^2}^2 \leq -2(\lambda_A - C_Y - L_{\text{Lip}}) \|\Psi^\perp(t)\|_{L^2}^2,$$

which integrates to the exponential estimate consistent with (3.5).

C Riesz projector bounds and base potentials

For each n , let $A_n(y) \geq 0$ be the nonnegative vertical operator defined in Section 2 and fix a circle Γ of radius $\lambda^*/2$ centered at 0 inside the spectral gap (uniform in y). The fiberwise Riesz projector is

$$\Pi_n(y) = \frac{1}{2\pi i} \int_{\Gamma} (z - A_n(y))^{-1} dz.$$

Strong commutation gives the joint projector

$$\Pi_{\text{coh}}(y) = \Pi_1(y)\Pi_2(y)\Pi_3(y), \quad \|\Pi_{\text{coh}}\|_{L^2 \rightarrow L^2} \leq 1.$$

Differentiating with respect to a base coordinate y^α gives

$$\partial_\alpha \Pi_n(y) = \frac{1}{2\pi i} \int_{\Gamma} (z - A_n(y))^{-1} (\partial_\alpha A_n(y)) (z - A_n(y))^{-1} dz,$$

hence

$$\|\partial_\alpha \Pi_n(y)\|_{L^2 \rightarrow L^2} \leq \frac{C}{(\lambda^*)^2} \|\partial_\alpha A_n(y)\|_{H^2 \rightarrow L^2}.$$

Under bounded internal geometry and C^m base dependence, $\|\partial_\alpha A_n(y)\|_{H^2 \rightarrow L^2}$ is uniformly bounded, so $\|\partial_\alpha \Pi_{\text{coh}}(y)\|_{L^2 \rightarrow L^2} \leq C'_\Pi$ and

$$\|\Pi_{\text{coh}} \Psi\|_{H^1} \leq (1 + C'_\Pi) \|\Psi\|_{H^1}.$$

This is the standard mechanism behind Lemma 2.4; see [5, 3].

Base potentials. Adding a nonnegative confining potential $V(y)$ improves tail coercivity on Y^4 and underlies Darbo on noncompact bases, but does not change the fiber operators $A_n(y)$ and hence does not change λ^* or λ_A .

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Dependency and reproducibility statement

The arguments use the locally stated model, the cited FP I results, and standard mathematical literature. No numerical packet or later MTT paper is needed. The examples are analytic checks of hypotheses, not fitted physical data. Computational applications must cite this paper in the forward direction.